

Task 4

Pseudocode

OUTPUT "Enter quantity of products"

INPUT q

OUTPUT "Enter price per item"

INPUT p

OUTPUT "Enter discount percentage"

INPUT d

OUTPUT "Enter tax percentage"

INPUT t

~~IF q OR $p \leq 0$~~

IF q OR $p \leq 0$:

OUTPUT "Enter positive values"

$valid1 \leftarrow false$

ENDIF

IF d OR $t \geq 100$:

OUTPUT "percentage should be less than 100"

$valid2 \leftarrow FALSE$

ENDIF

$s \leftarrow q * p$
 $a \leftarrow s - (s * d) / 100$
 $\text{final-bill} \leftarrow a + (a * t) / 100$

IF valid1 = TRUE AND valid2 = TRUE:

Subtotal =

~~$s = q * p$~~

~~discounted amount~~

~~Subt~~

$s \leftarrow q * p$

~~$a \leftarrow \text{discounted amount} \leftarrow s - (s * d) / 100$~~

~~$\text{final-bill} \leftarrow a + (a * t) / 100$~~

$\text{final-bill} \leftarrow a + (a * t) / 100$

OUTPUT ~~q~~

OUTPUT("Subtotal =", s)

OUTPUT("Discounted amount =", a)

OUTPUT("Final Bill =", final-bill)

PAC

Given Data

Required results

q

s

p

a

d

final bill

t

valid1

valid2

Processing required

Alternative solutions

IF q or p ≤ 0

IF d or t ≥ 100

IF valid1 = TRUE AND valid2 = TRUE

Define outputs as content

Flowchart

Start

OUTPUT "Enter quantity of products"

INPUT q

OUTPUT "Enter price per item"

INPUT p

OUTPUT "Enter discount percentage"

INPUT d

OUTPUT "Enter tax percentage"

INPUT t

Is q or $p \leq 0$

Yes

OUTPUT "Enter positive values"

valid1 = FALSE

No

Is d or $t \geq 100$

Yes

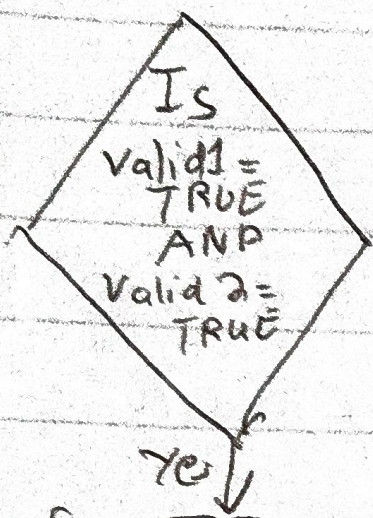
OUTPUT "Percentage should be less than 100"

valid2 = FALSE

No

End

(A)



$S = q * p$
 $a = s - (s * d)$
 $\text{final_bill} \leftarrow a + (a * t) / 100$

OUTPUT "Subtotal =", s

OUTPUT "Discounted amount =", a

OUTPUT "Final bill =", final_bill

END

Task 4

X IPO

INPUT	Processing	Module reference	Output
quantity of products (q)	Input Input p, q, d, t Validate p, q, d, t	Input validate	"Enter quantity of products"
discount percentage (d)	display error	Output	Enter price per item
price per item (p)	IF $q > 0$ or $p < 0$: valid1 $\leftarrow$ false valid2 $\leftarrow$ false	Condition assignment assignment	Enter discount percentage
tax (t)			Enter tax percentage